USBC Connector

J1 USB_C_Receptacle_USB2.0

VBUS ← Vbus ← D5 1N5819WS ← PTC1 MF-FSMF035X-2 ← Vin

C3 4.7u GND

CC1 A5 R5 5.1k GND

CC2 B5 R7 5.1k GND

D- A7 USB_D-

D- B7

D+ A6 USB_D+

D+ B6

SBU1 A8 NC

SBU2 B8 NC

SHIELD GND S1 A1

Vbus Vbus

R20 5.1k R22 5.1k

JP1 JP2

R24 5.1k R10 5.1k GND GND

Peripheral(Default)/Host Selection
Host Mode: Solder open bridges, cut shorted bridges
Peripheral: Default

3.3V

U9 NCP115AMX330TCG

Vin → IN, OUT → +3V3, EN, GND

C39 4.7u C41 4.7u

SJ6 SHORT +3V3_FPGA

FPGA Power Note:
The FPGA is normally powered off the +3V3 supply.
If you want to conserve power and turn off the FPGA,
cut the jumper shorting the +3V3 and the +3V3_FPGA
and supply the +3V3_FPGA externally.
You may use an external load switch deriving power
from the +3V3 and controlled by the RP2350 for example.

FPGA 1.2V Core

U1 NCP115AMX120TCG

+3V3_FPGA → IN, EN, GND, OUT → +1V2

C7 4.7u C37 4.7u

Peripheral Headers

J2 J3

	J2		J3	
iCE Flash	ICE_12 | IO2	ICE_RST	ICE_SSN	ICE_SO
	ICE_13 | IO3		ICE_SI	ICE_SCK
	+3V3_FPGA	+3V3_FPGA	SSn ICE_25	ICE_23 SSn
iCE PMOD	ICE_28	ICE_31	Tx ICE_19	ICE_18 Tx
	ICE_32	ICE_34	Rx ICE_27	ICE_26 Rx
	ICE_36	ICE_38	SCK ICE_21	ICE_20_G3 SCK
	ICE_42	ICE_43		
iCE Misc	ICE_37	ICE_35_G0	+3V3_FPGA	+3V3_FPGA
	ICE_11	ICE_6	ICE_DONE | ADC1	ICE_LED_R
	ICE_10 | PB	ICE_9	ADC2 | PB	ICE_LED_G
	VIO_BANK_2	VIO_BANK_2	SSn PICO_PMOD_A1	ICE_LED_B
iCE PMOD	ICE_48	ICE_47	Tx PICO_PMOD_A2	PICO_PMOD_B1 SSn
	ICE_46	ICE_45	Rx PICO_PMOD_A3	PICO_PMOD_B2 Tx
	ICE_44_G6	ICE_2	SCK PICO_PMOD_A4	PICO_PMOD_B3 Rx
	ICE_3	ICE_4		PICO_PMOD_B4 SCK
BOOT | ADC	SDA_GPIO2	RESET	+3V3	+3V3
	SCL_GPIO3	USB_BOOT	Vbus ADC3	ADC4 Vin
	ADC7		ADC5 | VREF	ADC6
	GND		GND	

iCE Prog
iCE–Pico PMOD
LED/ADC
Pico PMOD
ADC/Power

USBC

Raspberry Pico

+3V3 → C11 4.7u, C10 0.1u, C12 0.1u, C15 0.1u, C16 0.1u, C17 0.1u, C18 0.1u, C19 0.1u → GND

+1V1 → C4 0.1u, C5 0.1u, C6 4.7u → GND

Vin → C20 4.7u, C38 4.7u, L1 3.3u → VREG_LX → GND

+3V3 → R26 22 → VREG_AVDD → C23 4.7u → GND

C35 4.7u, C9 0.1u → GND

+3V3 → R25 100

Place R4/R5 close to RP2040

USB_DP 67 ← USB_P ← R4 22 ← USB_D+
USB_DM 66 ← USB_N ← R5 22 ← USB_D-

DVDD 23	QSPI_DVDD 69	IOVDD 10	USB_OTP_VDD 59
DVDD 22		IOVDD 50	ADC_AVDD 58
DVDD 21		IOVDD 43	
DVDD 10		IOVDD 36	
		IOVDD 24	
		IOVDD 15	
		IOVDD 9	
		IOVDD 1	

VREG_AVDD 61
VREG_VIN 64
VREG_FB 65
VREG_LX 63
VREG_PGND 62

QSPI_SS 75
QSPI_SD0 72
QSPI_SD1 74
QSPI_SD2 73
QSPI_SD3 70
QSPI_SCLK 71

XIN 30
XOUT 31

RUN 35

SWCLK 33
SWD 34

MISO0	GPIO0	77	LED_G
CSn0	GPIO1	78	LED_R
SCK0	GPIO2	79	SDA_GPIO2
MOSI0	GPIO3	80	SCL_GPIO3
MISO0	GPIO4	1	ICE_SI
CSn0	GPIO5	2	ICE_SSN
SCK0	GPIO6	3	R9 22 ICE_SCK
MOSI0	GPIO7	4	ICE_SO
MISO1	GPIO8	6	SRAM_SS
SSn1	GPIO9	7	LED_B
SCK1	GPIO10	8	GPIO10
MOSI1	GPIO11	9	GPIO11
MISO1	GPIO12	11	Lane0_N
SSn1	GPIO13	12	Lane0_P
SCK1	GPIO14	13	Lane1_N
MOSI1	GPIO15	14	Lane1_P
MISO0	GPIO16	15	Clock_N
CSn0	GPIO17	16	Clock_P
SCK0	GPIO18	18	Lane2_N
MOSI0	GPIO19	19	Lane2_P
MISO0	GPIO20	20	ICE_27
CSn0	GPIO21	21	R6 22 ICE_35_G0 FPGA Clock
SCK0	GPIO22	22	R34 22 ICE_20_G3 Suggested data clock
MOSI0	GPIO23	23	ICE_19
MISO1	GPIO24	25	ICE_26
SSn1	GPIO25	26	ICE_23
SCK1	GPIO26	27	ICE_21
MOSI1	GPIO27	28	ICE_18
MISO1	GPIO28	36	ICE_9
SSn1	GPIO29	37	ICE_11
SCK1	GPIO30	38	ICE_25
MOSI1	GPIO31	39	ICE_RST
MISO0	GPIO32	40	PICO_PMOD_A3
CSn0	GPIO33	42	PICO_PMOD_A1
SCK0	GPIO34	43	PICO_PMOD_A4
MOSI0	GPIO35	44	PICO_PMOD_A2
MISO1	GPIO36	45	PICO_PMOD_B3
SSn1	GPIO37	46	PICO_PMOD_B1
SCK1	GPIO38	47	PICO_PMOD_B4
MOSI1	GPIO39	48	PICO_PMOD_B2
GPIO40_ADC0		49	ICE_DONE | ADC0
GPIO41_ADC1		52	ADC1
GPIO42_ADC2		53	ADC2 | PB
GPIO43_ADC3		54	ADC3
GPIO44_ADC4		55	ADC4
GPIO45_ADC5		56	ADC5 | VREF
GPIO46_ADC6		57	ADC6
GPIO47_ADC7		58	

Raspberry Pi
RP2350B QFN-80

U3 RP2350_80QFN

GND 81

https://datasheets.raspberrypi.com/rp2350/hardware-design-with-rp2350.pdf

ABM8-272-T3 Y1 12MHz
C21 10p C1 10p GND GND
R1 1k +3V3
R16 10K RESET GND

JST Debug Connector
J5 BM03B-SRSS-TB
SWCLK SWD GND
NPTH

R7 10K Hold FPGA in reset while the pico is coming up

+3V3 → R30 0402 10K, R31 0402 10K → ADC5 | VREF, ADC6

Note:
Set the GPIO high to enable power to the
reference when the ADC needs to be calibrated.
Remove resistors to use the ADC on the external pins.

+3V3 → D7 1N5819WS → R12 1K (Cathode current) > 0.5mA → Vref = 2.5V → U4 TL431DBZ, C36 0.1u → GND

R29 2.2K ADC7 → R28 10K GND

Power monitor:
Adding R29 makes it possible to connect directly
to the positive terminal of a nominally 3.6V
lithium battery. R29 is 2.2K so that at fully
charged battery at about 4V yields 3.3V on GPIO47_ADC7.
An external resistor in series with R29 allows for
higher battery voltages. R29 acts as a current
limiting resistor if ADC7 is misused.
e.g. connected directly to the positive terminal of a 12V battery

HSTX FPC Connector

22 pin Top Contact 0.5mm FPC

+3V3 → R23 10K, R13 10K

+3V3		22	
SDA_GPIO2		21	
SCL_GPIO3		20	
ICE_19		18	
ICE_27		17	
GPIO	GPIO11	15	
GPIO	GPIO10	14	
GPIO19	Lane2_P	12	SFV22R-2STE1HLF
GPIO18	Lane2_N	11	J6
GPIO17	Clock_P	9	
GPIO16	Clock_N	8	
GPIO15	Lane1_P	6	
GPIO14	Lane1_N	5	
GPIO13	Lane0_P	3	
GPIO12	Lane0_N	2	
GND		1	

FPGA Banks

+3V3_FPGA → SJ5 SHORT → VIO_BANK_2

Bank 0 (U6A ICE40UP5K-SG48ITR, VCCIO_0)

IOT_36b	25	ICE_25
IOT_37a	23	ICE_23
IOT_38b	27	ICE_27
IOT_39a	26	ICE_26
IOT_41a	28	ICE_28
IOT_42b	31	ICE_31
IOT_43a	32	ICE_32
IOT_44b	34	ICE_34
IOT_45a_G1	37	ICE_37
IOT_46b_G0	35	ICE_35_G0
IOT_48b	36	ICE_36
IOT_49a	43	ICE_43
IOT_50b	38	ICE_38
IOT_51a	42	ICE_42
RGB0	39	ICE_LED_G
RGB1	40	ICE_LED_B
RGB2	41	ICE_LED_R

Note:
Output-only when the pll
is used for 2 output mode

Bank 2 (U6C ICE40UP5K-SG48ITR, VCCIO_2)

IOB_0a	46	ICE_46
IOB_2a	47	ICE_47
IOB_3b_G6	44	ICE_44_G6
IOB_4a	48	ICE_48
IOB_5b	45	ICE_45
IOB_6a	2	ICE_2
IOB_8a	4	ICE_4
IOB_9b	3	ICE_3

Note:
Bank 2 is selectable IO voltage!
To select non–3.3V, cut the shorted jumper
and supply external voltage (1.8–3.3V) on
VIO_BANK_2.

Bank 1 is 3.3V IO!

Bank 1 (U6B ICE40UP5K-SG48ITR, SPI_VCCIO1)

ICE_DONE | ADC0 → SJ3 SHORT → GND

CDONE		CRESET	8 ICE_RST
IOB_13b			6 ICE_6
IOB_16a			9 ICE_9
IOB_18a			10 ICE_10 | PB
IOB_20a			11 ICE_11
IOB_22a			12 ICE_12 | IO2
IOB_23b			21 ICE_21
IOB_24a			13 ICE_13 | IO3
IOB_25b_G3			20 ICE_20_G3
IOB_29b			19 ICE_19
IOB_31b			18 ICE_18
IOB_32a_SPI_SO			14 ICE_SO
IOB_33b_SPI_SI			17 ICE_SI
IOB_34a_SPI_SCK			15 ICE_SCK
IOB_35b_SPI_SS			16 ICE_SSN

FPGA Power

+1V2 → R11 100 → SJ4 SHORT (Core current measurement) → +1.2V_CORE

VCC_PLL → C33 0.1u, C34 4.7u → GND

+1.2V_CORE → C27 0.1u, C40 0.1u → GND

U6D ICE40UP5K-SG48ITR: VCCPLL 29, VCC 5, VCC 30, VPP_2V5 24, GND 49

+3V3_FPGA → D1 1N4148WT-7 → C28 0.1u, C31 4.7u → GND

FPGA Flash

+3V3_FPGA → C26 0.1u → GND

U8 W25Q32JVSSIQ

ICE_SSN → R8 10K → CS 1
ICE_SCK → CLK 6
DI(IO0) 5 ← ICE_SO
DO(IO1) 2 → ICE_SI
IO2 3 → ICE_12 | IO2
IO3 7 → ICE_13 | IO3
VCC 8, GND 4

RN1B 10K, RN1A 10K, RN1C 10K, RN1D 10K → GND

FPGA LED

+3V3_FPGA → SJ2 SHORT → C13 4.7u → GND

D4 MHPA1010RGBDT

R 2 ← ICE_LED_R
G 3 ← ICE_LED_G
B 4 ← ICE_LED_B

FPGA Button

+3V3_FPGA → SW2 SW_Push → R21 10K ICE_10 | PB, C14 0.1u → GND

FPGA Done LED

+3V3_FPGA → R14 2.2K → ICE_DONE | ADC0 → D3 LTST-C191KGKT → GND

Pico Flash

+3V3 → C8 0.1u → GND

Note:
Add DNF if the Flash is
replaced with an SRAM.

SRAM_SS → R3 DNF, R27 DNF

USB_BOOT → D6 1N5819WS → R2 1K → QSPI_SS

U2 W25Q32JVSSIQ

QSPI_SS	CS	1	
QSPI_SCLK	CLK	6	
VCC		8	
GND		4	
DI(IO0)		5	QSPI_SD0
DO(IO1)		2	QSPI_SD1
IO2		3	QSPI_SD2
IO3		7	QSPI_SD3

RP2 LED

+3V3 → SJ1 SHORT → C32 4.7u → GND

D2 MHPA1010RGBDT

R 2 → R17 1k LED_R
G 3 → R18 1k LED_G
B 4 → R19 1k LED_B

PICO Button

+3V3 → R32 10K → USB_BOOT

SW1 BOOT → GND

The button functions as a boot select
as well as GPIO.

R33 5.1k ADC2 | PB

SRAM

+3V3 → C42 0.1u → GND

R15 10K SRAM_SS

U5 APS6404L-3SQN-SN (substituted APS6404L-3SQR-SN)

SRAM_SS	CS	1	
QSPI_SCLK	CLK	6	
VCC		8	
GND		4	
DI(IO0)		5	QSPI_SD0
DO(IO1)		2	QSPI_SD1
IO2		3	QSPI_SD2
IO3		7	QSPI_SD3

LOGO4 OSHW ID: US002372

LOGO2 WEEE Logo

FID1 FID2 FID3 FID4 FID5 FID6 Fiducial

H1 H2 H3 H4 MountingHole

WWU
Rob Frohne
tinyVision.ai Inc.
Sheet: /
File: pico2-ice.kicad_sch

Title: PICO2–ICE

Size: A2	Date: 2025-07-23	**Rev: 0.21**
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